SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 2.1: Properties of Waves — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Properties of Waves
Driving Question	DRIVING QUESTION: "Why does the matatu's horn change pitch as it passes, and why does the FM radio signal disappear inside the Kabete tunnel — and what do these two mysteries reveal about how ALL waves behave?"

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon Launch — "The Matatu Horn and the Fading Radio"**	Students listened to an audio clip of a matatu horn Doppler-shifting as it passed and noted an FM radio signal cutting out inside the Kabete tunnel. They recorded their initial noticings and wonderings on sticky notes for the class Driving Question Board (DQB). Key teacher move: cold-call at least 4 students to share observations before accepting any explanations — enforce the observe/explain distinction explicitly.	Sound and radio signals are both wave phenomena. Waves transfer energy from one place to another without permanently displacing the medium through which they travel (e.g., a rope wave moves energy along the rope, but the rope itself does not travel). The two anchoring phenomena — the pitch shift and the signal loss — are not unrelated accidents; they are symptoms of universal wave behaviour that this sub-strand will unpack systematically.	The matatu horn pitch-shift is caused by the Doppler effect: relative motion between source and observer compresses or stretches the wave pattern, changing the perceived frequency. The FM signal loss inside the Kabete tunnel is caused by a combination of reflection, interference, and poor diffraction of short-wavelength FM waves around the tunnel walls. Both phenomena will be fully explained only after all wave properties have been studied — making them ideal anchoring phenomena. Remind students to return to these two mysteries at the end of every lesson to update the DQB.
Lesson 2 **What Is a Wave? Building Our First Model from Everyday Kenyan Phenomena**	Students generated waves on a long slinky (or knotted rope) stretched across the classroom floor, alternately shaking one end transversely and compressing-releasing it longitudinally. They also observed ripples spreading across a basin of water (or a video of Lake Victoria surface waves). A coloured ribbon tied to the middle of the slinky was watched carefully: it oscillated in place rather than travelling with	A wave is a disturbance that transfers energy from one point to another without net transfer of matter. Transverse waves (e.g., water ripples, electromagnetic waves including the FM signal from Classic 105 Nairobi) have particle oscillation perpendicular to the direction of wave travel, and display identifiable crests, troughs, and amplitude. Longitudinal waves (e.g., the matatu horn sound) have particle	The ribbon demonstration is the conceptual anchor: because the ribbon stays near its equilibrium position while the wave energy travels forward, students can see directly that matter is not transported. This directly challenges the common misconception that 'the wave carries the air along.' For the DQB: the matatu horn produces a longitudinal sound wave; the FM radio signal is a transverse electromagnetic wave

	the wave. Teacher should prompt: 'Watch the ribbon — where does it go?'	oscillation parallel to wave travel, with compressions and rarefactions instead of crests and troughs.	— both obey the same fundamental definition. Pedagogical note: insist students draw labelled diagrams of both wave types before moving on; diagram accuracy predicts exam performance on this topic.
Lesson 3 **The Wave Equation: Connecting Speed, Frequency, and Wavelength**	Students used a ripple tank (or PhET 'Wave on a String' simulation) to observe that increasing the frequency of the source while the medium remains unchanged causes the wavelength to decrease proportionally, while the wave speed stays constant. They recorded frequency (f), wavelength (λ), and speed (v) in a table and computed v/f and v/λ for each row. Teacher should circulate and ask: 'What do you notice about the product f × λ each time?'	The wave equation $v = f\lambda$ expresses that wave speed equals frequency multiplied by wavelength. For a given medium, wave speed is fixed; therefore frequency and wavelength are inversely proportional. For FM radio waves (electromagnetic) travelling at the speed of light ($c \approx 3 \times 10^8$ m/s), Classic 105 FM (105 MHz) has wavelength $\lambda = 3\times10^8 / 105\times10^6 \approx 2.86$ m. For sound in air at ~340 m/s, a 340 Hz matatu horn has wavelength λ = 1 m.	The equation $v = f\lambda$ is the mathematical spine of the entire sub-strand. Every subsequent phenomenon — diffraction, resonance, Doppler — references this relationship. Pedagogical note: require students to show full substitution and unit analysis in every calculation; dropped units are the single largest source of lost marks in KCSE wave questions. Connect to DQB: the specific wavelengths of FM vs. AM radio are why they behave differently in the Kabete tunnel — a preview students should record as a 'partial answer' on the DQB.
Lesson 4 **Wave Types and Properties: Transverse, Longitudinal, and the Language of Waves**	Students completed a card-sort activity matching wave vocabulary (amplitude, period, frequency, wavelength, crest, trough, compression, rarefaction, wave speed) to labelled diagrams of both transverse and longitudinal waves. They also measured the period of a pendulum (a simple accessible oscillator) by timing 20 complete oscillations and dividing, then computed frequency. Teacher: cold-call 3 students to explain the difference between amplitude and wavelength in their own words before whole-class correction.	Key wave vocabulary with precise definitions: Amplitude (A) — maximum displacement from equilibrium, measured in metres; determines energy carried. Period (T) — time for one complete oscillation (seconds); $T = 1/f$. Frequency (f) — number of complete waves per second (Hz); $f = 1/T$. Wavelength (λ) — distance between two successive identical points on a wave (e.g., crest to crest), in metres. Wave speed (v) — distance travelled per unit time (m/s). Compressions and rarefactions replace crests and troughs in longitudinal waves.	Precise vocabulary is non-negotiable in wave physics — examiners penalise conflation of amplitude with wavelength, or period with frequency. The pendulum activity grounds abstract definitions in a visible, measurable oscillator familiar from everyday Kenyan life (e.g., a stone on a string). DQB connection: the matatu horn's pitch is its frequency; the 'loudness' of the horn is its amplitude. These two properties are independent — a critical conceptual distinction that resolves a common student confusion about why a distant matatu sounds quieter (lower amplitude, same frequency) but not lower-pitched.
Lesson 5 **The Wave Equation: Connecting Frequency, Wavelength, and Speed ($v = f\lambda$)**	Students derived $v = f\lambda$ from first principles: if f waves pass a point per second, and each wave is λ metres long, then in one second the wave front has advanced f × λ metres — which is the definition of speed. They then applied the equation to three Kenyan-context problems: (1) calculating the wavelength of a 98.4 MHz Capital FM signal in air; (2) calculating the frequency of a sound wave of wavelength 0.5 m in air; (3) calculating the speed of a wave on Lake Victoria given f = 2 Hz and λ = 3 m.	$v = f\lambda$ is derived, not merely memorised — students who derive it forget it less often. The equation applies universally to all wave types (mechanical and electromagnetic) but the value of v differs: electromagnetic waves in vacuum travel at $c = 3 \times 10^8$ m/s; sound in air at ~340 m/s at room temperature; water waves much slower. Frequency is determined by the source and does not change when a wave moves from one medium to another; wavelength and speed both change.	The derivation-first approach is a deliberate NGSS Science and Engineering Practice (Using Mathematics and Computational Thinking). Pedagogical note: this lesson intentionally revisits $v = f\lambda$ from Lesson 3 with greater algebraic rigour and Kenyan radio-station data. Students who struggled in Lesson 3 often consolidate here. DQB update: students can now calculate that FM wavelengths (~3 m) are much shorter than AM wavelengths (~300 m) — a quantitative clue to why FM fails in the Kabete tunnel

	Teacher: wait 3 minutes of silent working before accepting any answers.		while AM survives.
Lesson 6 **Diffraction of Waves: Why AM Survives the Kabete Tunnel**	Students observed diffraction using a ripple tank (or video/simulation): straight water waves passing through gaps of varying width. They recorded that when the gap width ≈ wavelength, the waves spread widely and evenly into the shadow region; when the gap width >> wavelength, most energy continues straight ahead with only slight edge-bending. They also compared a long-wavelength rope wave passing around a large book vs. a short-wavelength rope wave hitting the same obstacle.	Diffraction is the spreading of waves when they pass through a gap or around an obstacle. The key rule: diffraction is most pronounced when the gap width (or obstacle size) is approximately equal to the wavelength (gap ≈ λ). AM radio waves (wavelength ~300 m) diffract easily around the ~10 m Kabete tunnel opening; FM radio waves (wavelength ~3 m) do not diffract appreciably around the same opening and so are blocked. This is why AM radio works in tunnels and FM does not — directly answering half of the driving question.	This is the lesson where the Kabete tunnel mystery is first fully resolved. Teacher: make this moment explicit — return to the DQB and physically move the FM/tunnel sticky note from the 'Questions' column to the 'Explained' column. Pedagogical note: students frequently confuse diffraction (single gap/obstacle) with interference (two sources/gaps). Establish the distinction now. Exam tip: KCSE questions on diffraction almost always ask students to state the condition for maximum diffraction and to draw the resulting wavefront pattern — both should be practised here.
Lesson 7 **Interference of Waves: When Waves Meet, Add, and Cancel**	Students used a ripple tank with two point sources vibrating in phase (or a double-slit water wave simulation) and observed alternating regions of large disturbance (antinodal lines) and calm water (nodal lines). They also listened to a teacher-produced audio demo: two speakers playing the same tone — walking between them revealed loud and quiet zones. The spatial pattern of nodes and antinodes was sketched and labelled.	When two waves meet, the resultant displacement at any point is the algebraic sum of the individual displacements — this is the Principle of Superposition. Constructive interference occurs when crests meet crests (or troughs meet troughs): path difference = nλ, resultant amplitude is maximum. Destructive interference occurs when crests meet troughs: path difference = (n + ½)λ, resultant amplitude is zero (cancellation). These two outcomes produce the stable nodal/antinodal pattern seen in the ripple tank.	Interference is a wave-exclusive behaviour — particles cannot cancel each other. This makes interference the definitive evidence that something is a wave. Pedagogical note: the algebraic sum rule is the conceptual core; students who understand superposition can derive both constructive and destructive conditions rather than memorising them. DQB connection: interference partially explains the patchy FM signal inside the Kabete tunnel — reflected waves from the tunnel walls interfere with the direct signal, creating pockets of cancellation. Encourage students to update the DQB with this added nuance.
Lesson 8 **Standing Waves: When a Wave Meets Itself**	Students vibrated a stretched string (or rubber tubing) at increasing frequencies using a signal generator and vibration generator (or by hand), observing the formation of stable loop patterns at specific resonant frequencies. They counted nodes (points of permanent zero displacement) and antinodes (points of maximum displacement) for the fundamental, second harmonic, and third harmonic. A video of a Nairobi street guitarist plucking strings of different lengths was used as a real-world parallel.	Standing (stationary) waves form when two waves of equal frequency and amplitude travelling in opposite directions superpose in the same medium. Nodes are fixed points of permanent destructive interference; antinodes are fixed points of maximum constructive interference. The distance between two adjacent nodes = λ/2. The fundamental (first harmonic) has one antinode; the nth harmonic has n antinodes. Standing waves do not transfer energy along the medium — energy is stored between nodes.	Standing waves are a special, structured consequence of superposition — they are not a new phenomenon but an elegant outcome of interference in a bounded medium. Pedagogical note: insist that students can identify and count nodes and antinodes from a diagram; this is a consistent KCSE examination skill. The guitarist analogy grounds the physics in an everyday Kenyan context: shorter string = higher fundamental frequency = higher note, directly applying v = fλ and L = λ/2 for the fundamental. DQB link: standing waves

			inside the Kabete tunnel explain why some frequencies are amplified and others cancelled — a preview of Lesson 11.
Lesson 9 **Diffraction and Interference: Why Waves Bend Around Obstacles and Cancel or Reinforce Each Other**	Students completed a consolidation practical combining diffraction and interference: they directed a torch (or laser pointer, if available) through a single narrow slit cut in card (diffraction) and then through a double slit (interference), projecting the pattern onto a white wall. They sketched and compared the two distinct patterns. A second activity used a ripple tank to revisit the two-source interference pattern from Lesson 7 and explicitly linked nodal lines to destructive interference path-difference calculations.	Diffraction (single gap/obstacle) and interference (two or more coherent sources) are distinct but related wave behaviours, both explained by the Principle of Superposition. Diffraction spreads a wave; interference between the diffracted waves from a double slit creates the regularly spaced bright and dark fringes (antinodal and nodal lines). Key quantitative relationship for double-slit interference: path difference for constructive interference = $n\lambda$; for destructive = $(n + ½)\lambda$. Both phenomena require wave nature — they cannot be explained by particle models.	This consolidation lesson is deliberately placed after separate treatments of diffraction (Lesson 6) and interference (Lesson 7) to give students time to build each concept before combining them. Pedagogical note: the most common student error is applying the diffraction condition (gap $\approx \lambda$) to interference scenarios and vice versa — use targeted questioning to check this. The torch/laser double-slit demonstration is high-impact and low-cost; it is worth repeating slowly with extended wait time for student observation. DQB update: the combined diffraction-plus-interference model is now sufficient to fully explain the patchwork FM signal inside the Kabete tunnel.
Lesson 10 **Bending Light: Refraction, Snell's Law, and Total Internal Reflection**	Students directed a ray of light (ray box or torch with slit) from air into a glass block and a water trough, measuring the angle of incidence (θ_i) and angle of refraction (θ_r) at the boundary. They tabulated $\sin\theta_i$ and $\sin\theta_r$ for at least five angles and computed the ratio $\sin\theta_i / \sin\theta_r$ for each row, observing that the ratio is constant. The demo was extended to show total internal reflection when the ray travels from glass to air beyond the critical angle — illustrated with a fibre-optic strand (or a flexible clear tube filled with water).	Refraction is the change in wave speed (and direction) when a wave passes from one medium to another. Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$, where n is the refractive index of the medium. The refractive index $n = c / v$ (speed of light in vacuum / speed of light in the medium); $n > 1$ for all optical media. Total Internal Reflection (TIR) occurs when a wave travels from a denser to a less dense medium at an angle of incidence greater than the critical angle θ_c, where $\sin\theta_c = n_2/n_1$. TIR is the principle behind fibre-optic cables used in Kenya's national broadband network (NOFBI).	Refraction extends the wave model beyond sound and radio to light, demonstrating the universality that the driving question promises. Pedagogical note: the constant ratio (Snell's Law) should be discovered by students from their own data before being named — this is the NGSS practice of Analysing and Interpreting Data. The NOFBI fibre-optic context is highly relevant: Kenya's undersea fibre-optic cables (SEACOM, TEAMS) deliver internet to Mombasa using TIR. DQB link: refraction does not explain the Kabete tunnel directly, but it demonstrates that all waves — sound, light, radio — change speed at boundaries, which is a unifying principle to add to the board.
Lesson 11 **Resonance and Stationary Waves — Why the Tunnel Amplifies Some Sounds and Kills Others**	Students blew across the tops of test tubes of varying water levels to produce notes of different pitch, identifying the resonant lengths at which the sound is loudest. They also clapped once inside a long corridor (or watched a video of clapping inside the Kabete tunnel) and listened for the echo pattern. The teacher used a tuning fork held	Resonance occurs when a system is driven at its natural frequency, causing it to oscillate with maximum amplitude. In an enclosed space (such as the Kabete tunnel), standing waves form at specific frequencies that satisfy the boundary conditions: for a tunnel open at both ends, resonant frequencies occur at $f = nv/2L$ (n =	This lesson brings standing waves (Lesson 8) and interference (Lesson 7) together in the specific context of the Kabete tunnel, making it a critical synthesis point. Pedagogical note: the test-tube resonance activity is accessible, requires no specialist equipment, and provides vivid qualitative evidence of resonance — do not skip it

	near a resonant air column (Kundt's tube or a tube closed at one end) to show resonance at specific lengths matching L = λ/4 (fundamental, closed pipe) and L = λ/2 (fundamental, open pipe).	1, 2, 3…). Frequencies matching these conditions are amplified (antinodal condition); frequencies that produce nodal conditions are attenuated or cancelled. This explains why some sounds are louder inside the tunnel and why certain FM frequencies suffer more signal loss than others at specific positions.	even if time is tight. The quantitative condition (f = nv/2L) should be applied to a sample tunnel length (~200 m) to compute which frequencies resonate — this calculation directly answers a component of the driving question. Update the DQB: the tunnel mystery is now fully explained by the combined action of diffraction (Lesson 6), interference (Lesson 7), and resonance (this lesson).
Lesson 12 **Final Explanation — Pulling It All Together: From the Matatu Horn to the Kabete Tunnel**	Students reviewed the completed Driving Question Board, now populated with evidence, explanations, and model diagrams accumulated over eleven lessons. They constructed a final annotated explanatory diagram (individually or in pairs) showing: (1) the matatu Doppler shift labelled with wave compression/stretching, frequency, and relative motion; (2) the Kabete tunnel signal loss labelled with FM wavelength, diffraction condition, interference, and resonance. The class shared and critiqued each other's diagrams using a structured peer-review protocol.	ALL wave behaviours studied in this sub-strand — rectilinear propagation, reflection, refraction, diffraction, interference, superposition, standing waves, resonance, and the Doppler effect — are manifestations of a single underlying reality: waves are energy disturbances governed by v = fλ and the Principle of Superposition. The matatu horn pitch-shift is the Doppler effect: motion compresses wave fronts ahead of the source (higher f, higher pitch) and stretches them behind (lower f, lower pitch). The Kabete tunnel FM signal loss is the combined result of: poor diffraction (FM λ ≈ 3 m << tunnel opening), destructive interference between direct and wall-reflected signals, and standing-wave nodal conditions at certain frequencies and positions.	This final lesson is the payoff of the entire sub-strand and should be treated as a celebration of cumulative understanding. Pedagogical note: the peer-review diagram activity is both a formative assessment and a consolidation strategy — circulate and look specifically for students who can articulate the causal chain (wavelength → diffraction condition → interference pattern → resonance) rather than listing facts in isolation. The driving question is now fully answered, and students should be able to explain both phenomena to a peer who was not in the class. Exam preparation note: KCSE wave questions frequently ask students to 'explain using the wave model' — the habit of linking every claim to a wave property (f, λ, v, amplitude, superposition) is the highest-yield exam skill to reinforce in this lesson.